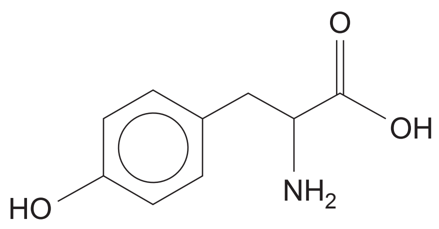
Examiner only

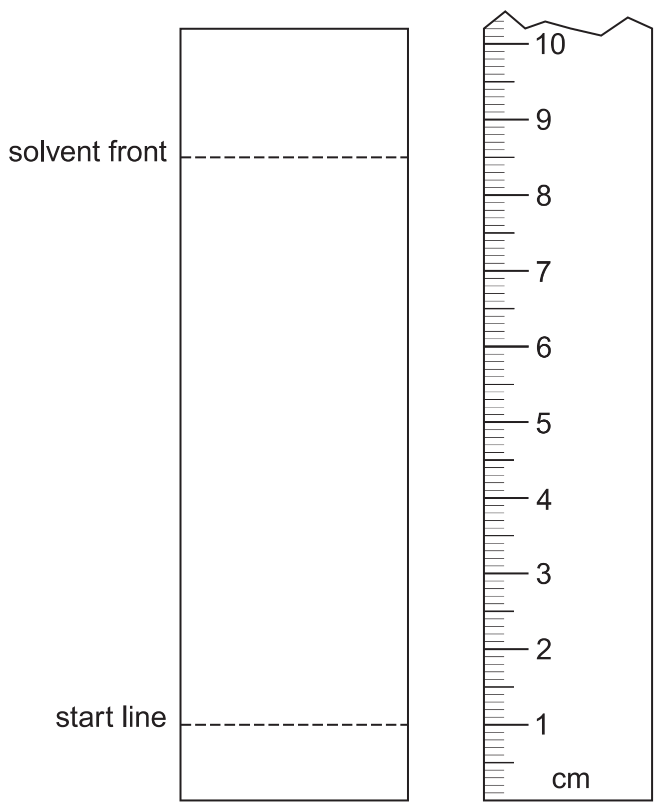
11. (a) Tyrosine is one of the amino acids making up casein, the protein in milk.



tyrosine

(i) Using a particular solvent tyrosine has an R_f value of 0.67.

On the chromatogram below show the spot given by tyrosine. [1]



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(ii) Write the formula for the zwitterion of tyrosine. [1]

(iii) Tyrosine is described as a hydrophobic amino acid, as its solubility in water is very low.

Use the formula of tyrosine to give a reason for this low solubility. [1]

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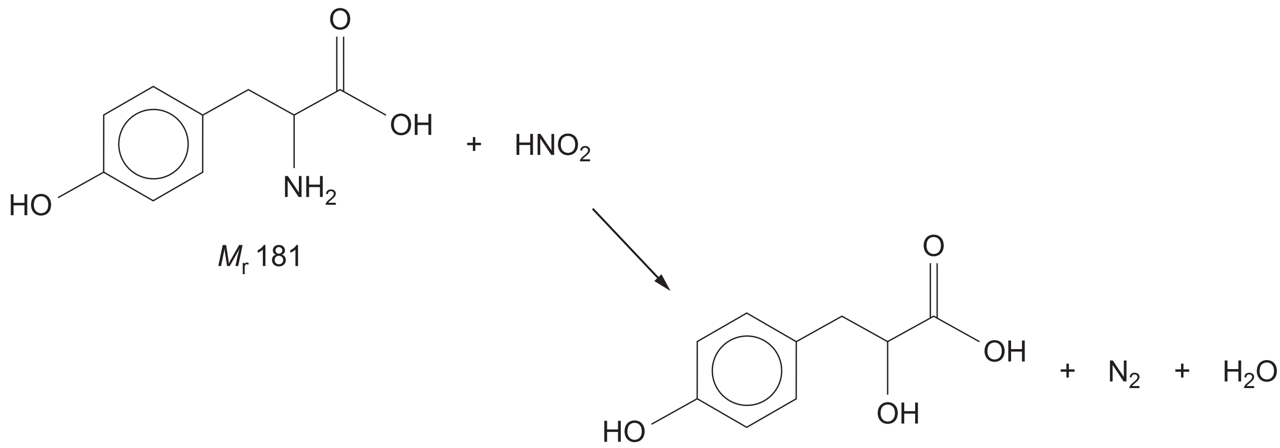
(iv) Amino acids, such as tyrosine, react with nitric(III) acid to give the corresponding hydroxy acid and nitrogen gas.



M_r 181

I. A sample of tyrosine was treated with an excess of nitric(III) acid and produced 147 cm^3 of nitrogen, measured at 298 K and 1 atm pressure.

Show that this volume of nitrogen will be produced from 1.09 g of tyrosine. [2]

II. The experiment was repeated with a sample of tyrosine from a different batch.
This time the same starting mass gave 132 cm^3 of nitrogen under the same conditions.

Suggest **one** reason for this different result. [1]

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(b) There is increasing interest in the production of important chemicals by biotechnology, rather than from the use of fossil fuels.

One of these compounds is butane-1,4-dioic acid, which can be made from glucose.

$$\underset{\substack{\text{glucose}\\ M_r\ 180}}{C_6H_{12}O_6} \xrightarrow{\text{anaerobic bacteria}} \underset{\substack{\text{butane-1,4-dioic acid}\\ M_r\ 118}}{HOOC(CH_2)_2COOH} + \text{other products}$$

In practice, a constant pH is maintained by the addition of calcium hydroxide.
As butane-1,4-dioate ions are produced they react with calcium hydroxide to give insoluble calcium butane-1,4-dioate, which is then filtered from the mixture.

$$\underset{\substack{\text{glucose}\\ M_r\ 180}}{C_6H_{12}O_6} + \underset{M_r\ 74}{Ca(OH)_2} \longrightarrow \underset{\text{calcium butane-1,4-dioate}}{(CH_2)_2(COO)_2Ca} + \text{other products}$$

(i) Calculate the atom economy for the production of calcium butane-1,4-dioate. [2]

Atom economy = ... %

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(ii) In an experiment, 54.0 g of glucose produced 41.2 g of calcium butane-1,4-dioate.

Calculate the minimum volume of sulfuric acid, of concentration 2.5 mol dm^{-3}, necessary to convert all the calcium butane-1,4-dioate into butane-1,4-dioic acid. [2]

$$(CH_2)_2(COO)_2Ca + H_2SO_4 \longrightarrow HOOC(CH_2)_2COOH + CaSO_4$$

Volume of sulfuric acid = ... cm^3

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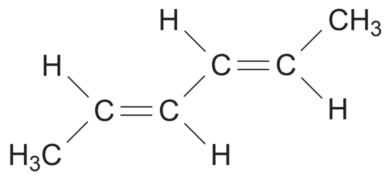
(c) Hexa-2,4-diene is one of the important products that can be produced from butane-1,4-dioic acid.



hexa-2,4-diene

Use the formula to help you to describe the ^{13}C NMR spectrum of this compound.

Your answer should include the number of peaks seen and your reasoning.
The position and size of the peaks is not required. [2]

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